

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B Tech- AI & DS	Date:	25-08-2026

Code of Conduct

I Y Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Greeshma Reddy

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Coreference Resolution

a) Referring expressions and possible antecedents

Referring Expression	Possible Antecedents
He	John, Mary
She	Mary, John
it	a ball, park, dog
him	John, Mary, dog
they	John, Mary, dog

b) Applying constraints

1. Gender and number agreement

- **He** → **John** because John is male.
- **She** → **Mary** because Mary is female.
- **it** → **ball** because a ball is singular and non-human.
- **him** → **dog** because *dog* is the most suitable singular masculine/animate antecedent in this context.
- **they** → **John + Mary + dog** because *they* is plural.

2. Recency

- Among compatible candidates, prefer the most recently mentioned entity.
- For **him**, *dog* is mentioned immediately before “him,” so dog is preferred.

3. Semantic compatibility

- A **ball** can be played with → **it** = **ball**.
- A **dog** can chase someone → **him** = **John** is also semantically possible.
- However, grammatical gender gives **him** → **John**, while “dog” has no inherent grammatical gender. Thus, **him** → **John** is the stronger resolution.

4. Coherence

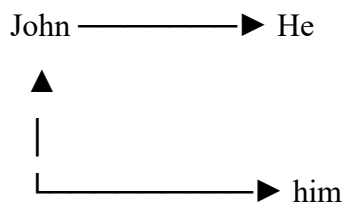
- The entity chains should remain consistent throughout the paragraph.
- John, Mary, ball, and dog form the main discourse entities.

c) Constraint graph/table

Mention	Domain	Gender/Number	Recency	Semantic Compatibility	Final Link
He	John, Mary	John ✓	John ✓	John ✓	John
She	John, Mary	Mary ✓	Mary ✓	Mary ✓	Mary

Mention	Domain	Gender/Number	Recency	Semantic Compatibility	Final Link
it	ball, park, dog	Singular/non-human	Ball ✓	Ball ✓	ball
him	John, Mary, dog	John ✓	Dog is recent	John = chased by dog ✓	John
they	John + Mary + dog	Plural ✓	Group is salient	All went home ✓	John + Mary + dog

A simple graph is:



Mary —————> She

Ball —————> it

John + Mary + Dog —————> they

d) Final coreference chains

- **John** → **He** → **him** → **they**
- **Mary** → **She** → **they**
- **ball** → **it**
- **dog** → **they**

Resolved paragraph:

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

e) Priority order of constraints

A suitable priority order is:

1. Gender/number agreement → 2. Semantic compatibility → 3. Coherence → 4. Recency

- **Agreement** is highly reliable because it eliminates grammatically impossible candidates.
- **Semantic compatibility** ensures that the reference makes sense in the situation.
- **Coherence** maintains consistent entity chains across the paragraph.
- **Recency** is useful as a tie-breaker, but it can sometimes produce an incorrect answer.

If **recency is relaxed**, the system can select an older but more semantically and grammatically appropriate

antecedent. For example, although *dog* is closer to **him**, **John** is a better antecedent because the dog is the one doing the chasing and John is the person being chased.

2. Conversation History and Dialog Act

a) Discourse planning steps

Before generating the response:

1. **Identify the user's situation:** The user has an important exam tomorrow and cannot concentrate.
2. **Determine the dialog act:** Provide **advice + encouragement**.
3. **Identify entities:** Maintain *exam*, *concentrate*, and *you*.
4. **Select discourse relation:** Use **Cause** → **Effect** or **Elaboration**.
5. **Choose required keywords:** Include at least two of **focus**, **break**, **confident**.
6. **Generate a polite and positive 2–3 sentence response.**
7. **Check consistency:** Advice should be realistic and encouraging.

b) Three possible responses

Response 1:

Since your exam is tomorrow, take a short break and return to studying with a clear focus. You have prepared for this, so stay calm and confident that you can do your best.

Response 2:

If you are unable to concentrate, take a short break and then study in small focused sessions for your exam. This can improve your focus and help you feel more confident.

Response 3:

Your exam is important, so do not worry if you cannot concentrate continuously; take a short break and restart calmly. With a clear focus and steady effort, you can feel more confident about the exam.

c) Evaluation and best response

Criteria	Response 1	Response 2	Response 3
2–3 sentences	✓	✓	✓
Advice + encouragement	✓	✓	✓
Exam coherence	✓	✓	✓
Focus/break/confident	✓	✓	✓
Cause/Effect or Elaboration	✓	✓	✓
Positive tone	✓	✓	✓

Best response: Response 1

It is the most natural because it directly connects the **exam** to the need for a **break and focus**, while the second sentence provides clear **encouragement**. It also uses all three useful keywords—**break, focus, confident**—without sounding forced.

d) Effect of violating two constraints

Violation of entity coherence: If the response suddenly talks about unrelated entities instead of the **exam** and **concentration**, the conversation becomes less relevant and the user may feel that the response does not understand their situation.

Violation of the 2–3 sentence constraint: A very long response could overwhelm a student who is already struggling to concentrate. A very short response might also fail to provide enough advice and encouragement.

3. Word Sense Disambiguation and Discourse

a) Resolve “bank”

The correct meaning of **bank** is **riverbank**, not financial institution.

Reason:

- The phrase “**by the river**” strongly indicates a geographical riverbank.
- The verb “**flooded**” is semantically compatible with a riverbank.
- A financial bank does not normally flood in the intended meaning of this sentence.
- Therefore:

bank → **riverbank**

b) Predicate logic representation

Let:

- $\text{River}(r) = r$ is a river
- $\text{Bank}(b) = b$ is a bank/riverbank
- $\text{Storm}(s) = s$ is a storm
- $\text{Action}(a) = a$ is a quick action
- $\text{location}(b,r) = \text{bank } b \text{ is located by river } r$
- $\text{flood}(b) = \text{bank } b \text{ flooded}$
- $\text{after}(\text{flood}(b),s) = \text{flooding occurred after storm } s$
- $\text{saved}(b,a) = \text{bank } b \text{ was saved by action } a$
- $\text{quick}(a) = \text{action } a \text{ was quick}$

A suitable representation is:

$\exists b \exists r \exists s \exists a [$

Bank(b) $\wedge$ River(r) $\wedge$ Storm(s) $\wedge$ Action(a)

$\wedge$ location(b,r)

$\wedge$ flood(b)

$\wedge$ after(flood(b), s)

$\wedge$ quick(a)

$\wedge$ saved(b,a)

]

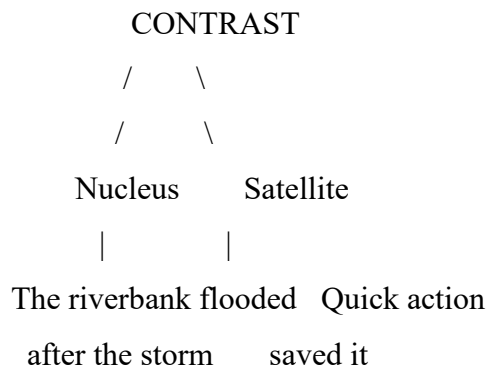
This preserves the important entities and relations.

c) Simple English paraphrase

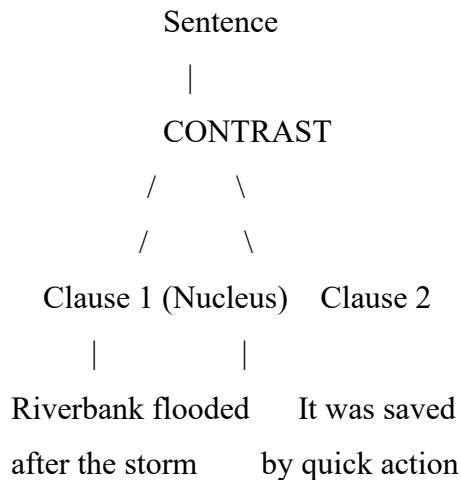
The riverbank near the river flooded after the storm, but quick action saved the riverbank.

This removes the ambiguity of **bank** while preserving the original meaning and contrast.

d) Simple RST-style discourse tree



More explicitly:



The word “**but**” signals the **Contrast** relation between the two clauses.

e) Advantages of constraint-based approach over pure statistical MT

A **constraint-based approach** is advantageous because it explicitly checks linguistic requirements rather than relying only on learned probability.

1. **Better word-sense accuracy:** Context such as *river* and *flooded* helps select **riverbank** instead of financial bank.
2. **Preserves important relations:** Predicate constraints ensure that the relation between the bank,

river, storm, flooding, and quick action is retained.

3. **Maintains discourse structure:** The **Contrast** relation indicated by *but* can be explicitly preserved.
4. **Reduces semantic errors:** Constraints prevent logically unsuitable interpretations.
5. **Improves coherence:** The same entity is maintained across both clauses instead of accidentally switching meanings.

Thus, constraint-based processing is particularly useful when **semantic correctness, reference resolution, and discourse structure** are more important than simply producing the most statistically probable translation.